MIL-T-9560B(USAF)
3 APRIL 1968
SUPERSEDING
MIL-T-9560A(USAF)
17 November 1959

MILITARY SPECIFICATION

TEST SET, RADIO FREQUENCY POWER AN/USM-68( )

1. SCOPE

1.1 This specification covers one type of general purpose, ruggedized, test equipment for measuring radio frequency (rf) power, designated Test Set, Radio Frequency Power AN/USM-68( ), hereinafter referred to as "the test set."

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-C-3432	Cable And Wire, Electrical (Power And Control; Semi-Flexible, Flexible, And Extra-Flexible, 300 And 600 Volts)
MIL-C-3767	Connectors, Plug And Receptacle, Electrical, (Power, Bladed Type), General Specification For
MIL-T-4807	Tests, Vibration And Shock, Ground Electronic (General Requirements For)
MIL-N-7513	Nomenclature Assignment, Contractors Method For Obtaining
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification For
MIL-T-21200	Test Equipment For Use With Electronic And Fire Control Systems, General Specification For

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-170	Moisture Resistance Test Cycle For Ground Signal Equipment
MIL-STD-202	Test Methods For Electronic And Electrical Component Parts
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-785	Requirements For Reliability Program (For Systems And Equipment)
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS3106	Connector, Plug, Electric, Straight

* (Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

* 3.2 Components. The test set shall consist of the following:

Item No.	Quantity	Description	Requirement
1	1 each	Test Set, Radio Frequency Power TS-1339/USM-68	3.8.1.1
2	1 each	Bolometer, Radio Frequency DT-216/USM-68	3.8.2
3	1 each	Bolometer, Radio Frequency DT-214/USM-68	3.8.2

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Item No.	Quantity	Description	Requirement
4	6 each	Bolometer Element DT-215/USM-68	3.8.3
5	6 each	Bolometer Element DT-217/USM-68	3.8.4
6	1 each	Attenuator, Fixed CN-588/U	3.8.5
7	1 each	Attenuator, Fixed CN-589/USM-68	3.8.5
8	1 each	Attenuator, Fixed CN-590/USM-68	3.8.5
9	1 each	Attenuator, Fixed CN-591/U	3.8.5
10	1 each	Adapter, Connector UG-1340/USM-68	3.8.6
11	1 each	Cord CG-92E/U (0 feet, 8 inches)	3.8.7
12	5 each	Calibration Chart	3.8.8
13	1 each	Combination Case	3.8.9

* 3.3 General specification. The requirements of MIL-T-21200 apply as requirements to this specification. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern. Exceptions and additions to the general specification shall be as follows.

* 3.3.1 Service conditions. The test set shall be designed and constructed to operate under any combination of service conditions specified in the general specification with exceptions and additions as follows.

* 3.3.1.1 Climatic.

a. Ambient temperature

(1) Operating: Minus 40° Centigrade (C) to plus 52°C

(2) Nonoperating: Minus 54°C to plus 71°C

b. The test set shall be designed and constructed to meet the minimum performance requirements during and after, exposure to the following environmental conditions, as detailed in Section 4 of this specification.

(1) Altitude: nonoperating - sea level to 50,000 feet.

(2) Salt atmosphere: as in coastal regions and during ocean transportation.

(3) Humidity

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* 3.3.1.2 Mechanical. The test set shall be designed and constructed to meet the following mechanical conditions, as specified in Section 4:

a. Vibration

b. Shock

* 3.3.1.3 Electrical. The test set shall operate from a 115 volt plus or minus 10 percent, 50 to 450 Hertz (Hz), single-phase alternating current (AC) power source in accordance with MIL-T-21200.

* 3.3.2 Reliability. The required degree of reliability in the design and construction of the equipment shall be determined by its operational requirements and shall be accomplished in accordance with MIL-STD-785 (see 6.2).

* 3.3.3 Lightweight material. Wherever practical, lightweight, high strength materials shall be used.

* 3.3.4 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's number.

* 3.3.5 Interference. The test set shall conform to the requirements of MIL-STD-461 governing generation or susceptibility to rf interference voltages and fields.

* 3.4 Nomenclature. Nomenclature shall be obtained in accordance with the requirements of MIL-N-7513. The parentheses appearing in the nomenclature will be dropped, or replaced by a letter to designate the particular design.

* 3.5 Design. The test set shall be constructed in accordance with the standard sample unless otherwise specified herein. The test set shall be a general purpose test equipment capable of measuring rf power within the frequency range from 500 megahertz (MHz) to 11,000 MHz. The rf power range shall be from 100 microwatts (uw) to 5 watts. Reliability of operation accessibility for service and ease of maintenance shall be of paramount importance in the construction of this equipment.

* 3.5.1 Frequency and power range. The test set shall be capable of measuring indications of power output over the following frequency and power range:

Frequency	Power
500 MHz to 11,000 MHz	100 uw to 10 milliwatts (mw)
500 MHz to 11,000 MHz	10 mw to 5 watts.

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* 3.6 Performance. The over-all accuracy of the test set, at the frequency and power range specified in 3.5.1, as indicated by a full scale deflection of the average power indicating meter, shall not be less than plus or minus 15 percent. For less than full scale deflection the accuracy shall not be less than plus or minus 18 percent. The accuracy shall be applicable when the test set is used in conjunction with appropriate bolometer, bolometer mount and attenuators. The operation and performance of the test set shall be based on the summation principle as applied to rf power measurements.

* 3.7 Details of components.

* 3.7.1 Test Set, Radio Frequency Power AN/USM-68( ). The test set shall consist of the following.

* 3.7.1.1 Test Set, Radio Frequency Power TS-1339/USM-68. The summation bridge shall be a direct-reading, watt-meter type bridge consisting of the following.

* 3.7.1.1.1 Power supply. The power supply shall be capable of operating from a power source as specified in 3.3.1.3. The unit shall rectify and filter this AC power source and supply the necessary regulated direct current (DC) power required for operating the equipment.

* 3.7.1.1.1.1 Direct current regulation. Following a 10 minute warm up period, the DC regulation shall be plus or minus 0.1 percent, within the specified input voltage variation limits, with other conditions remaining constant for all load requirements.

* 3.7.1.1.1.2 Root mean square ripple voltage. The root mean square (rms) ripple voltage shall be less than 0.1 percent of the measured DC output voltage.

* 3.7.1.1.1.3 Bias current. The bias current passing through the 200 ohm bolometer element, or an equivalent 200 ohm plus or minus 2 percent resistor attached to the interconnecting cable shall be 35 milliamperes (ma) plus or minus 1 ma in the 35 ma high power bolometer position and 4.0 ma plus or minus 0.2 ma in the 4.0 ma low power bolometer position.

* 3.7.1.1.1.3.1 Bias selector. A control shall be provided for the purpose of conveniently selecting the proper bias current for either of the specified bolometer elements. The control circuit shall be designed to preclude any possibility of applying improper bias current to either element when connected to (or in the process of connection) the bridge.

* 3.7.1.1.2 Step attenuator. The step attenuator shall provide attenuation to the metering circuit in steps of 0.2 decibel (db). Eleven positive steps and eleven negative steps shall be provided. When particular miniaturization techniques shall be used to minimize the physical size of the attenuator, the ohmic value of the resistors used in the attenuators shall be within plus or minus 0.1 percent of the rated value. The value of the resistors shall be

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determined by the following formula:

$$db = 20 \text{ Log}_{10} \frac{I_1}{I_2}$$

$$R = \frac{E}{I_2}$$

Where: I_1 = current with zero resistance inserted

I_2 = current with resistance inserted

E = regulated output voltage

R = resistance to be inserted (ohms)

* 3.7.1.1.3 Meters. Meters shall be in accordance with MIL-M-10304. The meters shall have a full scale value of 200 microamperes (ua) DC, and shall be hermetically sealed.

* 3.7.1.1.3.1 Balance meter. The balance meter shall zero at center scale. Scale markings shall be in accordance with the balance meter on the standard sample.

* 3.7.1.1.3.2 Power meter. The power indicating meter shall be a multirange meter with zero at the left. Two range scales shall be provided for relative power indication, and an additional similar scale, in a contrasting color, shall be included to indicate the calibration level for each power range. The 2 relative power indication scales shall have 50 scale divisions. One range scale shall be marked with numerals from 0 to 50, the other range scale shall be marked with numerals from 0 to 1.

* 3.7.1.1.4 Power cable. The AC power cable shall be 6 feet, 0 inch length of 3-conductor cable in accordance with MIL-C-3432, terminated at one end with a connector in accordance with MIL-C-3767/12A. The other end of the cable shall be permanently attached into the front panel of the bridge. The cable shall be securely attached to the front panel with a cord grip.

* 3.7.2 Bolometer mounts. Bolometer, Radio Frequency DT-216/USM-68 and Bolometer, Radio Frequency DT-214/USM-68 shall be designed and fabricated subject to the approval of the procuring activity from the standard sample of the rf bolometer mount in which the high power element and low element specified herein may be operated. The high and low power elements shall be noted to their respective mounts on one end with MS3106A10SL-3S and the other end with UG-21D/U. When the complete assembly, with bolometer installed, is connected to the test set it shall meet the performance requirements specified herein.

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* 3.7.2.1 Dimensions. The Bolometer, Radio Frequency DT-216/USM-68 over-all dimensions shall be 3 1/16 inches long by 2 inches in diameter and Bolometer, Radio Frequency DT-214/USM-68 over-all dimensions shall be 3 1/16 inches long by 2 inches in diameter.

* 3.7.3 Bolometer element DT-215/USM-68. The Bolometer Element DT-215/USM-68 (low power) shall be provided for use and tested in Radio Frequency Bolometer DT-216/USM-68. The bolometer element shall be of the Wollarston Wire Type, or equal, and when installed in the Radio Frequency Bolometer DT-216/USM-68 shall meet the following requirements for a minimum continuous operating period of one hour:

* a. The balance operational resistance when element is heated with 4.0 ma shall be 200 ohms (185 ohms approximately - cold)

b. Resistance tolerance plus 4 ohms to minus 6 ohms at 25°C

c. Differential resistance between sections 10 ohms

* d. Bias current 4.0 ma

e. Pulse repetition frequency (prf) per second 100

f. Pulse width 5 microseconds (usec)

g. Peak pulse power 2 watts

h. Energy per second (erg/sec) 100

i. Maximum Voltage Standing Wave Ratio of element and Radio Frequency Bolometer DT-216/USM-68, 1.35:1

* j. Configuration shall conform to common design illustrated in Figure 1.

* 3.7.4 Bolometer element DT-217/USM-68. The Bolometer Element DT-217/USM-68 (high power) shall be provided for use and tested in Radio Frequency Bolometer DT-214/USM-68. The bolometer element shall be of the carbon squadage type and when installed in the Radio Frequency Bolometer DT-214/USM-68 shall meet the following requirements for a minimum continuous operating period of one hour:

a. The balance operational resistance when element is heated with 35 ma shall be 200 (220 ohms approximately - cold).

b. Resistance tolerance plus 4 ohms to minus 6 ohms at 25°C'

c. Differential resistance between sections 10 ohms

d. Bias current 35 ma

e. PRF 100

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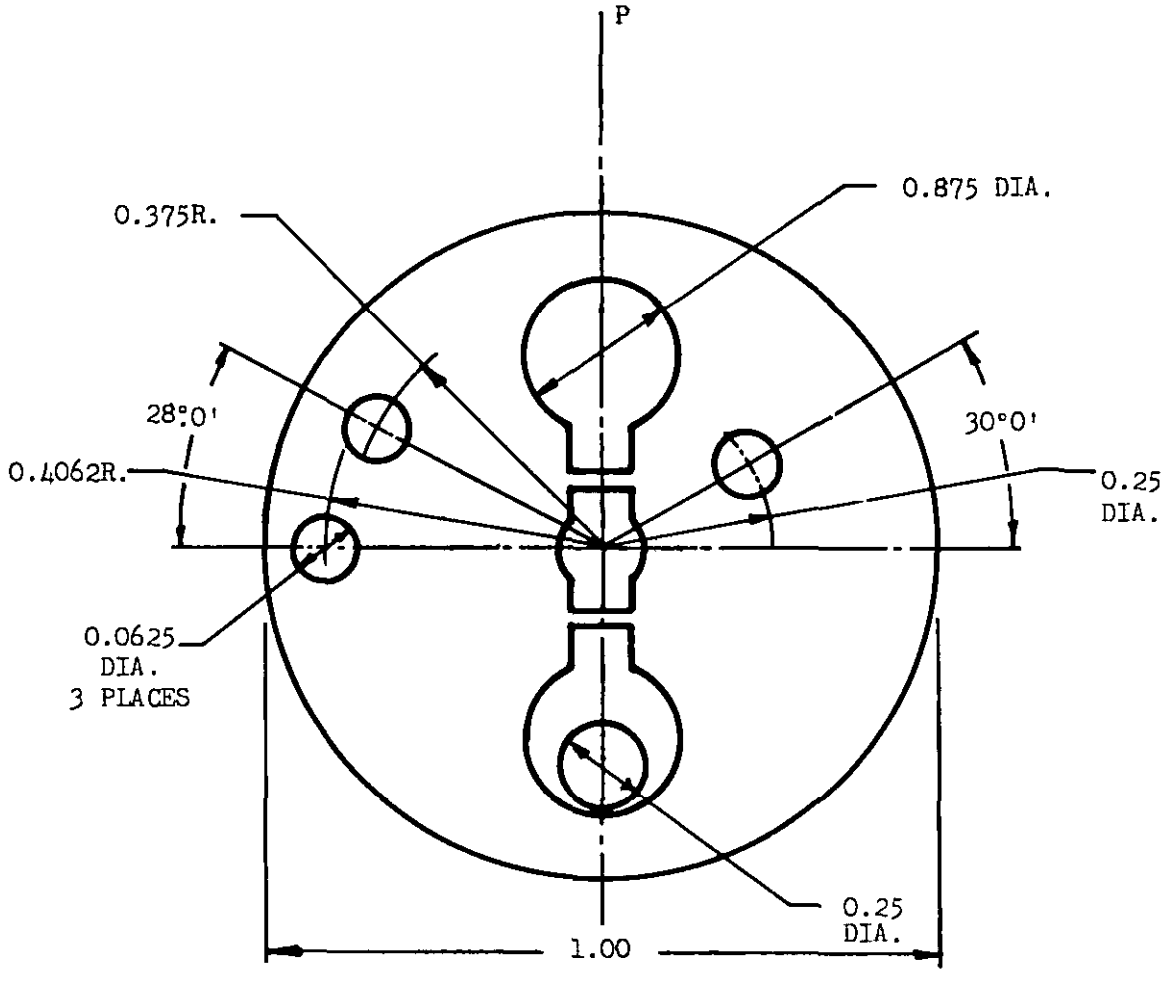


FIGURE 1. BOLOMETER ELEMENT COMMON DESIGN

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f. Pulse width 5 usec

g. Peak pulse power 200 watts

h. Energy 10,000 ergs/sec

i. Maximum voltage standing wave ratio (VSWR) of element and Radio Frequency Bolometer DT-214/USM-68, 1.35:1

* j. Configuration shall conform to common design illustrated in Figure 1.

3.7.5 Attenuators, fixed. Attenuator, Fixed CN-588/U; Attenuator, Fixed CN-589/USM-68; Attenuator, Fixed CN-590/USM-68 and Attenuator, Fixed CN-591/U shall be interchangeable with the standard sample attenuators. The attenuators and combination of attenuators shall be calibrated to the required accuracy of 0.2 db at each of the following frequencies: 500 MHz, 1300 MHz, 2500 MHz, 5800 MHz, 8500 MHz and 11,000 MHz. Each attenuator shall include the frequency range and nominal attenuation on the nameplate. The maximum attenuation deviation due to frequency sensitivity shall not exceed the specified over-all deviation limits over the entire frequency range. The attenuators and combination of attenuators shall meet the requirements specified in Table I.

* 3.7.6 Adapter, Connector UG-1340/USM-68. The Adapter, Connector UG-1340/USM-68 shall be as specified in 3.2. The VSWR shall not exceed 1.3:1 over the frequency range 500 MHz to 11,000 MHz. The adapter shall be calibrated for insertion loss at the following frequencies: 500, 1300, 2500, 5800, 8500 and 11,000 MHz. The maximum calibration error shall not exceed 0.05 db. The calibration figures obtained shall be permanently stamped or imprinted on the calibration chart.

* 3.7.7 Cord CG-92E/U (0 feet 8 inches). Cord CG-92E/U (0 feet, 8 inches) shall be fabricated from Radio Frequency Cable RG-9B/U terminated at each end with a Connector, Plug, Electrical UG-21D/U. The cord shall be accurately measured for attenuation at each of the following frequencies: 500 MHz, 1300 MHz, 2500 MHz, 8500 MHz and 11,000 MHz. Maximum attenuation shall not exceed 0.3 db.

* 3.7.8 Calibration charts. Calibration charts shall be furnished with the rf components of the test set. The charts shall give the calibration data and correction factor for each attenuator and combination of attenuators. The chart shall include the calibrated attenuation at each of the following calibration frequencies, to within 0.1 db: 500 MHz, 1300 MHz, 2500 MHz, 5800 MHz, 8500 MHz, and 11,000 MHz. The chart shall be sufficiently accurate and shall include any additional data necessary to obtain the value of attenuation to within 0.3 db at any intermediate frequency over the range of 500 MHz to 11,000 MHz.

* 3.7.9 Combination case. The components of the Test Set, Radio Frequency Power AN/USM-68( ) shall be enclosed in a combination case conforming to the requirements of MIL-T-21200.

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TABLE I

ATTENUATION CHARACTERISTICS

	CN-588	CN-589	CN-590	CN-591	CN-588 Plus CN-589	CN-590 Plus CN-591 and CN-588
Nominal Attenuation (db)	3	3	7	10	6	20
Maximum Frequency Sensitivity (db)	0.5	0.5	0.5	0.5	0.75	1.0
Over-all Deviation From Nominal (db)	±0.5	±0.5	±0.5	±0.75	±0.75	±1.0
PRF	100	100	100	100	100	100
Pulse Width (usec)	5	5	5	5	5	5
Peak Pulse Power (watts)	10,000	10,000	6000	2000	10,000	10,000
Maximum VSWR (From Either End)	1.3:1	1.3:1	1.3:1	1.3:1	1.3:1	1.3:1
Maximum Calibration Error (db)	0.2	0.2	0.2	0.2	0.2	0.2
Frequency Range (MHz)	500 to 11,000	500 to 11,000	500 to 11,000	500 to 11,000	500 to 11,000	500 to 11,000
Impedance (ohms)	50	50	50	50	50	50
Coaxial Size (inch)	3/8	3/8	3/8	3/8	3/8	3/8

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* 3.7.9.1 Cover, Test Set CW-534/USM-68. The case cover shall be secured to the case by 4 spring-loaded pull-down latches. The cover shall be detachable. The latches shall be mounted on the cover and hooks mounted on the case.

* 3.8 Color. The final paint coat shall be light gray enamel, color No. 16376, in accordance with FED-STD-595.

* 3.9 Identification of product. The physical item shall be marked in accordance with MIL-STD-130.

3.10 Government-loaned property. When the contract or purchase order so provides, the Government will loan the following to the contractor upon his request:

Item No.	Stock Number	Description	Quantity
1		Test Set, Radio Frequency Power AN/USM-68( )	1 each

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the test set shall be classified as follows:

a. Quality conformance inspection see 4.3

b. Preproduction tests see 4.6

* 4.3 Quality conformance inspection. Each test set shall be subjected to the quality conformance inspection which shall consist of the following tests as described under 4.5:

a. Mechanical inspection tests see 4.5.1

b. Electrical tests see 4.5.2

4.3.1 Additional tests. In addition, each test set shall be subjected to any other tests specified herein which the procuring activity considers necessary to determine compliance with the requirements of this specification.

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4.4 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.5 Test methods.

4.5.1 Mechanical inspection tests. The test set shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship are in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Completeness

b. Nameplates, identification markings and labels

c. Ease of operation of controls, adjustable and sliding parts, thumb screws, latches and switches

d. Finishes (paint and plating)

e. Check of solder joints

f. The fit of components in their respective positions

g. Check of lubrication and rust prevention

h. Loose fastening and securing devices or parts

i. Cable runs between components including plugs and receptacles

j. Spare parts, fuses, wrenches and specified accessories

k. Other visual defects

4.5.2 Electrical tests. The test set shall be given all electrical tests necessary to confirm that all circuits are inherently sound and in compliance with the requirements of this specification. The electrical tests shall include, but not necessarily be restricted to the following:

a. Over-all accuracy of the complete test set when measuring one millivolt (mv) of rf power at 500 MHz, 2500 MHz and 11,000 MHz.

b. Over-all accuracy of the complete test set when measuring 25 mv or 50 mv of rf power at 500 MHz, 2500 MHz and 11,000 MHz.

c. Attenuation accuracy of each rf attenuator and combination of attenuators indicated in Table I.

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4.5.2.1 Continuity. Each electrical component and each cord and cable shall be given a continuity test to ascertain if it is wired and connected correctly and that good electrical contact is obtained.

4.5.2.2 Operating voltages. The operating voltage at all important points shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be done with all controls set for normal operation of the equipment.

* 4.5.3 Direct current regulation. The test for DC regulation shall be conducted with the equipment as set up in Figure 2 and in accordance with the following test procedure. After 10 minutes warm-up, the power supply shall regulate within plus or minus 0.1 percent over the input voltage range of 115 volts plus or minus 10 volts:

a. Set Variac to 115 volts input to the Test Set, Radio Frequency Power AN/USM-68( ).

b. Balance Voltmeter LVM-5, (or equal). Note reading.

c. Set input voltage with Variac to 105 volts. Rebalance Voltmeter LVM-5. Note reading.

d. Subtract the readings in steps "b" and "c." Limit: 20 mv maximum.

e. Set input voltage with Variac to 125 volts. Rebalance Voltmeter LVM-5. Note reading.

f. Subtract the readings in steps "b" and "e." Limit: 20 mv maximum.

* 4.5.4 RMS ripple voltage. The test for rms ripple voltage shall be conducted with the equipment as set up in Figure 2 in accordance with the following procedure. Ripple voltage shall be less than 0.1 percent of measured DC output voltage:

a. Set input voltage to the Test Set, Radio Frequency Power AN/USM-68( ) at 115 volts.

b. Read ripple voltage on Ballantine type AC voltmeter, or equal. Limit: Less than 20 mv.

* 4.5.5 Bias current. The test for the DC bias current shall be conducted with the equipment as set up in Figure 3 in accordance with the following procedure:

a. With Test Set, Radio Frequency Power AN/USM-68( ) set for high power and line voltage set for 115 volts rms, adjust high bias control, (located through the front panel) for 7.00 volts, read on Voltmeter LVM-5.

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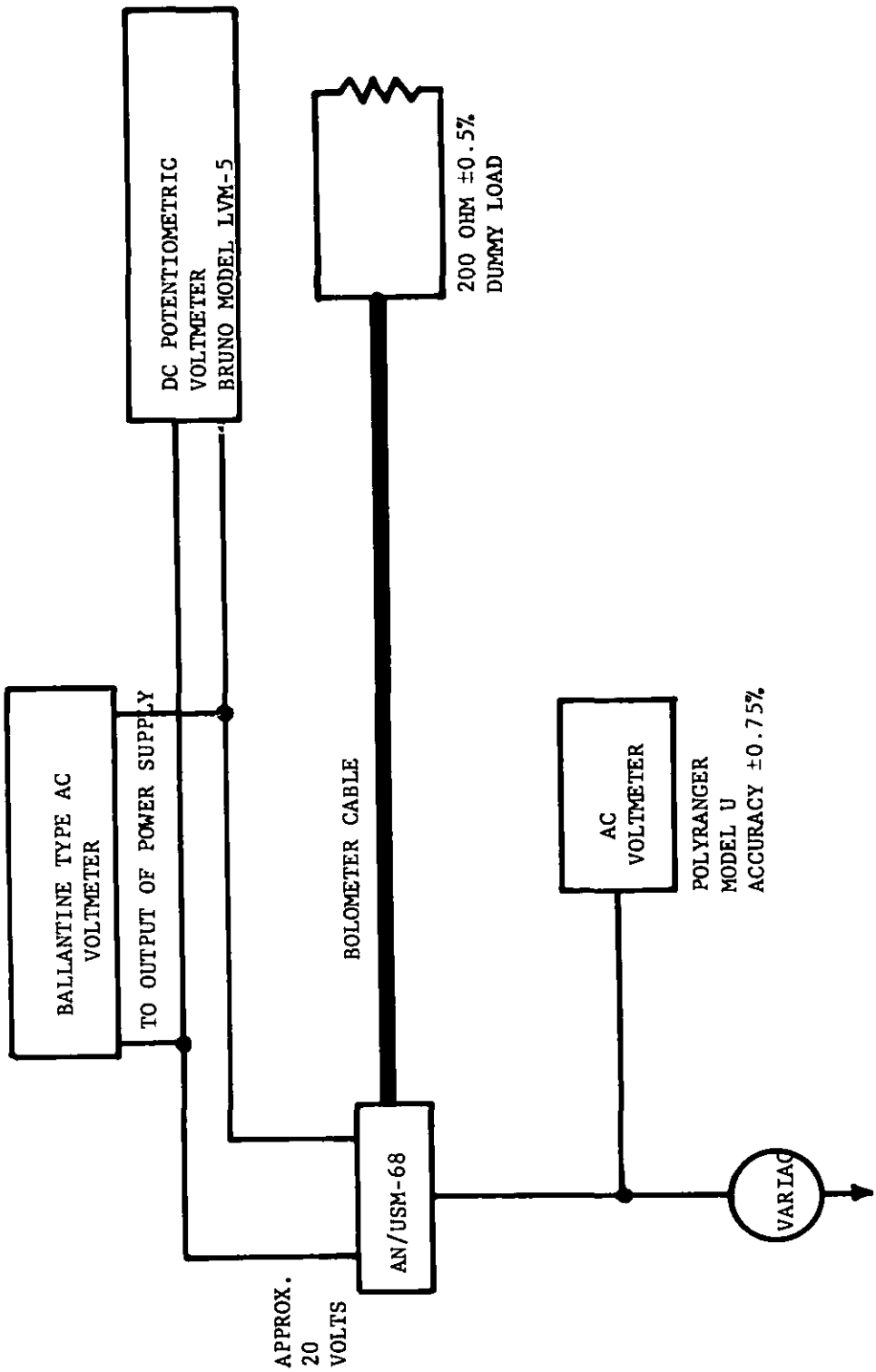


FIGURE 2. DC CURRENT REGULATION & RIPPLE TEST

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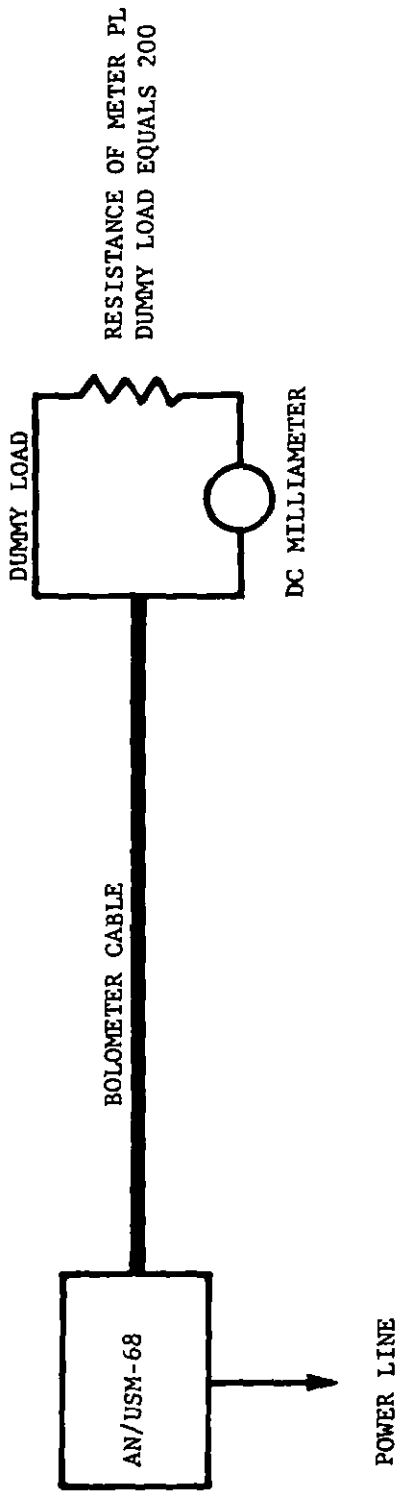


FIGURE 3. DC BIAS CURRENT TEST SET

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b. With Test Set, Radio Frequency Power AN/USM-68( ) set for low power and line voltage set for 115 volts rms, adjust low bias control (located through the front panel) for 0.90 volts, read on Voltmeter LVM-5.

* c. Replace set up shown in Figure 3 by that shown in Figure 4.

d. Set, Test Set, Radio Frequency Power AN/USM-68( ), to set ZERO in high power position. Note reading. Limit: 35 ma plus or minus 1 ma.

e. Repeat step "d" for low power position. Note reading. 4.5 ma plus or minus 0.2 ma.

4.5.6 Step attenuator. The step attenuators shall provide attenuation to the metering circuit in steps of 0.2 db. With the function switch set for calibration level, compensate attenuator set plus 2.2 db, rotate the compensate attenuator through all its positions. The readings on the block scale of the power meter shall change by approximately one division for each change of position of the compensate attenuator switch.

* 4.5.7 Bridge accuracy. Bridge accuracy shall be conducted with the equipment as set up in Figure 4 in accordance with the following procedure:

a. Set bridge in adjust balance position with the adjust balance control, maximum counter-clockwise position, compensate for zero db, and range switch set to high power. Note the voltage reading across 200 ohms dummy load resistor "E." Rotate adjust balance control so that the power meter reads 50 mv. Note the voltage reading across the 200 ohm dummy load resistor "Ep." Substitute the values for "E" and "Ep" in equation below. The value calculated shall be within plus or minus 6 percent of 50 mv.

$$\text{Substitution Power} = \frac{E^2 - Ep^{22}}{200}$$

b. Repeat step "a" for 40, 30, 20, 10 mv. In each case the calculated value shall not differ from the indicated value by more than plus or minus 6 percent of full scale.

c. Repeat steps "a" and "b" for the low power range. Levels of 1.0, 0.8, 0.6, 0.4, 0.2 mv.

* 4.5.8 Over-all accuracy measurement. Over-all accuracy shall be conducted with the equipment as set up in Figure 5 in accordance with the following procedure. Accuracy at full scale shall not be less than plus or minus 15 percent. For less than full scale, the accuracy shall not be less than plus or minus 18 percent:

a. Set power meter to high power range.

b. Set signal source for 50 mv at 4000 MHz using the standard power meter.

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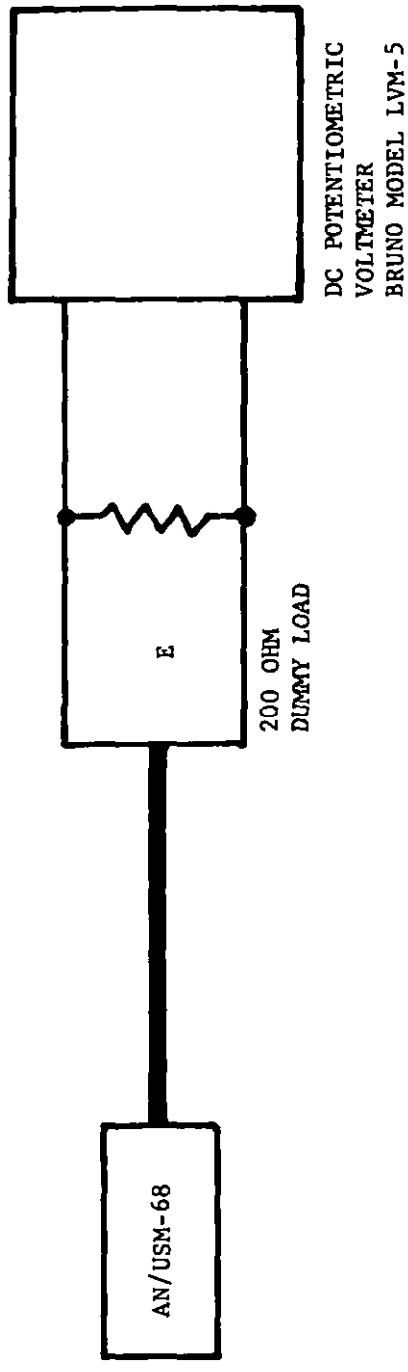


FIGURE 4. BRIDGE ACCURACY TEST SET

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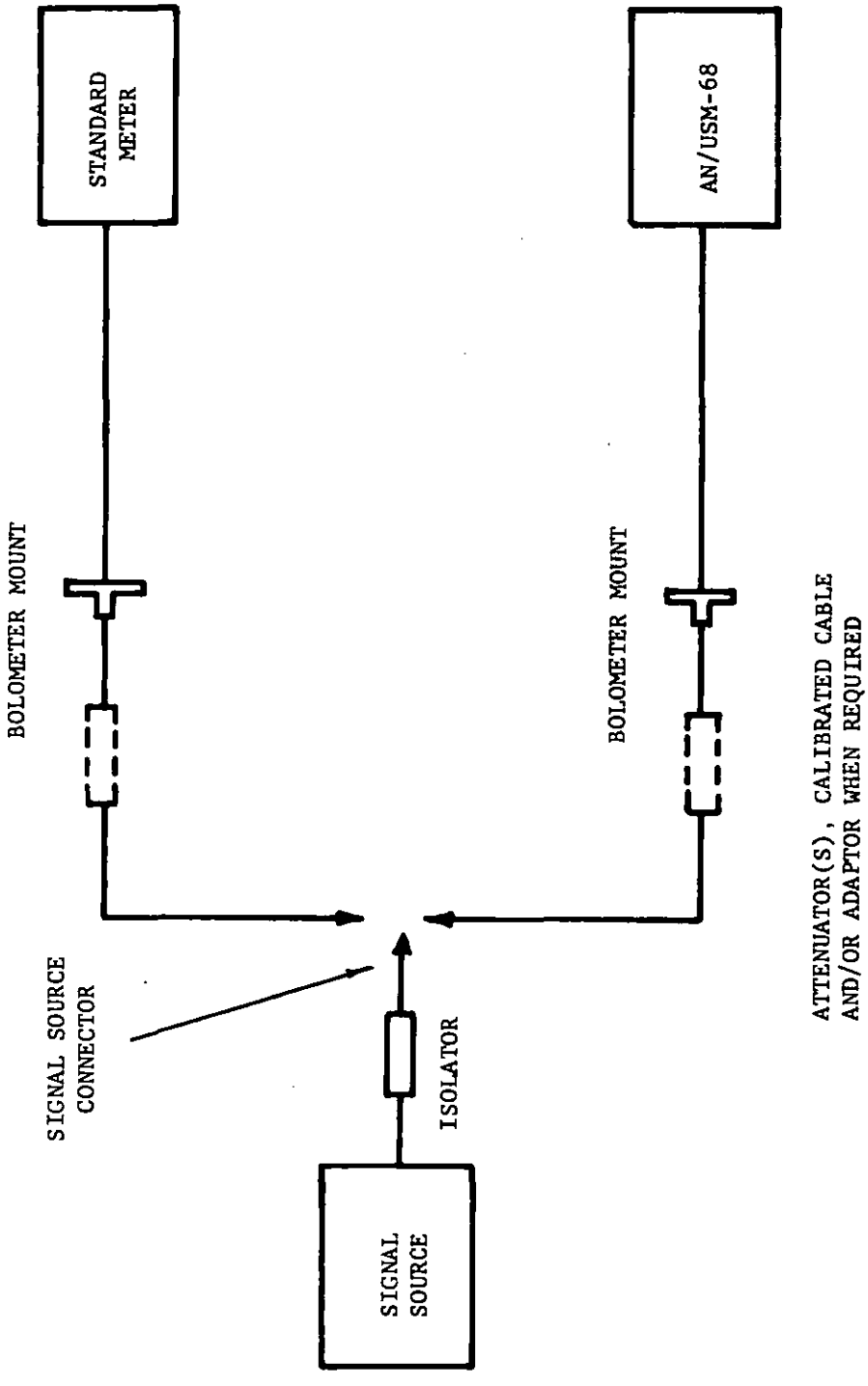


FIGURE 5. OVERALL ACCURACY TEST SET

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c. Transfer the signal source connectors to the test set and measure the signal level. Limit: 50 mw plus or minus 15 percent.

d. Should the test set read beyond full scale, set the standard test set to 40 mw and repeat step "c." Limit: 40 mw plus or minus 15 percent.

e. Set signal source for 25 mw using the standard power meter.

f. Transfer the signal source connector to the test set and measure the signal level. Limit: 25 mw plus or minus 18 percent.

g. Repeat step "a" through "f" for one mw and 0.5 mw level using the low power range of the test set in conjunction with the low power bolometer. Limit: 1 mw plus or minus 15 percent.

h. Repeat step "g" with the 20 db combination of attenuators and the right angle adapter, except with the input signal power level set at 100 mw and 50 mw. Limit: 1 mw plus or minus 15 percent for 100 mw input 0.5 mw plus or minus 18 percent for 50 mw input.

* 4.5.9 Measurement of characteristics of rf attenuators and right angle connector. The tests shall be conducted with the equipment as set up in Figures 6 and 7 in accordance with the following procedure in 4.5.9.3.

4.5.9.1 Requirements for rf attenuators (in combination for 6 db and 20 db).

Nominal Attenuator (db)	3	6	10	20
Maximum Frequency Sensitivity (db)	0.5	0.5	0.5	1.0
Over-all Deviation From Nominal (db)	0.5	0.5	0.75	1.0
Maximum VSWR	1.3:1	1.3:1	1.3:1	1.3:1

4.5.9.2 Requirements for right angle connector.

Attenuation	(Print calibrated value on calibration chart)
VSWR	(No greater than 1.3:1)

* 4.5.9.3 Test procedure for VSWR and attenuation (see Figure 7).

a. Set signal source to 11,000 MHz and balance the null indicator for zero.

b. Insert the component under test. Null indicator will deflect off zero.

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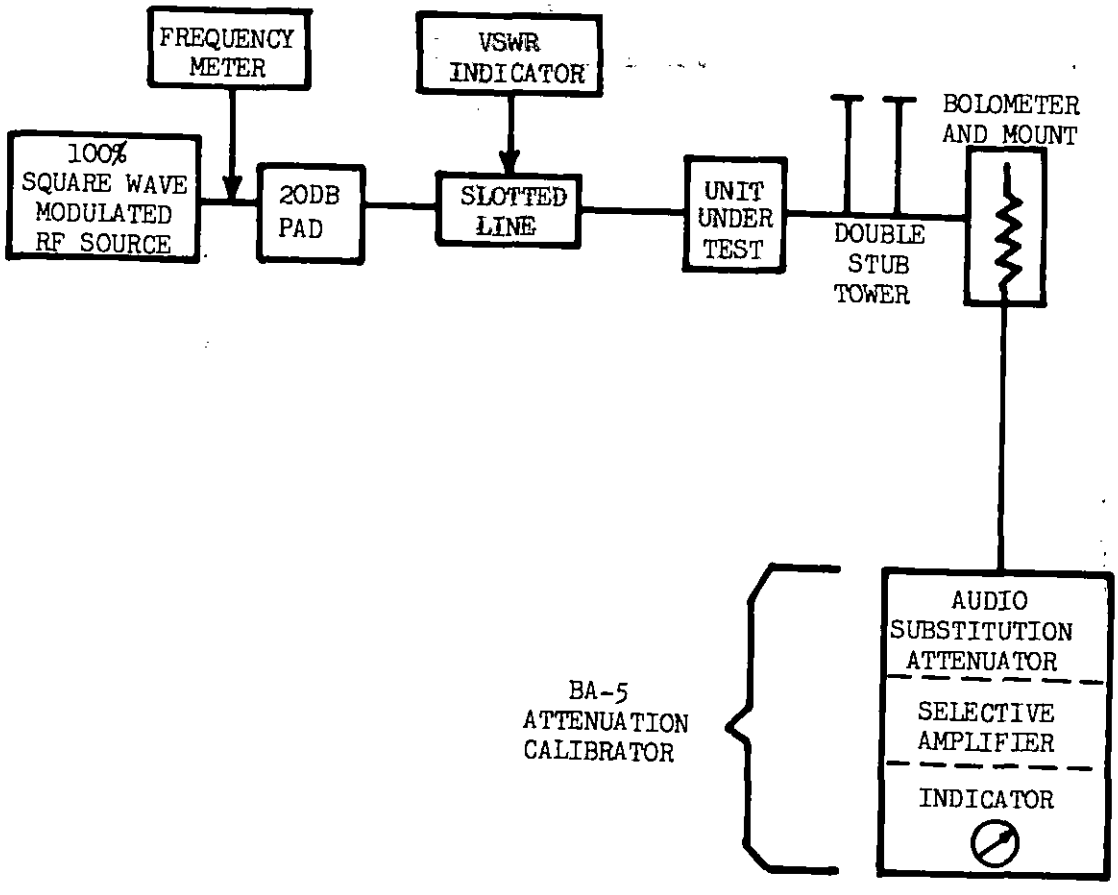


FIGURE 6. ATTENUATION AND VSWR

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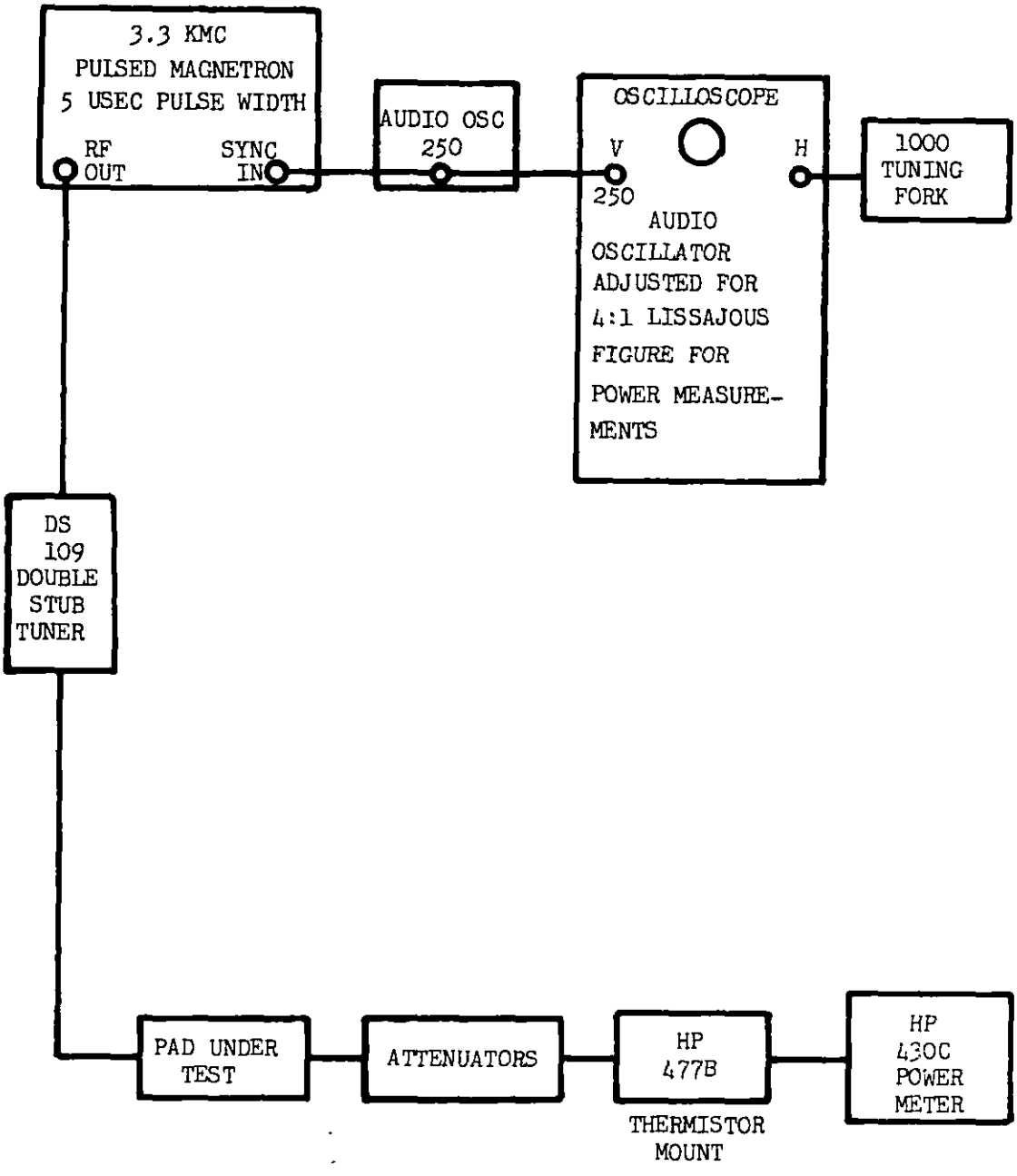


FIGURE 7. ATTENUATED PULSE POWER TEST.

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c. Reset attenuator calibration attenuation controls for balance on the null indicator meter. The numbers shown on attenuator calibration attenuator dials is the attenuation of the component under test.

d. Move probe along the slotted line and note the value of the VSWR.

e. Repeat steps "a" through "d" for 8500 MHz, 5800 MHz, 2500 MHz, 1300 MHz, and 500 MHz.

4.5.9.4 Requirements for pulse power. Each attenuator when properly terminated, shall meet the requirements listed above for the attenuators in combination, after subjection to the pulse power levels listed below:

Nominal attenuation (db)	3	7	10
Pulse repetition rate (pps)	100	100	100
Pulse width (usec)	5	5	5
Peak pulse power (watts)	10,000	6,000	10,000
Average power equivalent	5	3	5

* 4.5.9.5 Test procedure for pulse power (see Figure 6).

a. Set pulse modulated power source to the level shown above.

b. Connect attenuator as shown in Figure 6.

c. After the attenuator has been subjected to the pulse power requirement, repeat steps "a" through "d" of 4.5.9.3.

* 4.5.10 Bolometer and bolometer mount. The VSWR and resistance test shall be conducted with the equipment as set up in Figure 7 in accordance with the following procedures. The resistance with specified bias current applied shall be 200 ohms, plus 4 ohms minus 6 ohms. The VSWR shall not exceed 1.35:1.

a. With the proper bias current applied, apply rf power at 11,000 MHz and rebalance the test set.

b. Move probe along slotted line and note VSWR.

c. Repeat steps "a" and "b" for 8500, 5800, 2500, 1300 and 500 MHz.

d. Remove rf power, set function switch to "Set Zero" and reset zero.

e. Disconnect the bolometer mount without altering the set zero control and replace with a resistance decode box.

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f. Set the resistance decode dial for zero on the galvanometer. Note resistance value.

g. Repeat steps "a" through "f" for the low power bolometer element.

* 4.5.10.1 Pulse power test bolometer elements. The high and low power bolometer elements in their proper mounts shall meet the VSWR requirements after subjection to the pulse power levels listed below. This test shall be conducted with the equipment set up in Figure 6. Adjust the pulsed signal source to the values listed below:

	High Power Element	Low Power Element
Pulse repetition rate (pps)	100	100
Pulse width (usec)	5	5
Peak pulse power (watts)	200	2
Energy per second (ergs)	10,000	100

4.5.10.2 Test procedure.

a. Set pulse modulated power source to the level shown above.

b. Connect bolometer mount as shown.

c. Repeat steps "a" and "b" as specified in 4.5.10. Pulsed power shall not adversely affect the VSWR as indicated prior to pulse power test.

* 4.5.11 Test equipment. The following test equipment, or equal, is required to perform the tests in accordance with the set ups in Figures 2 through 7.

Item	Component	Type and Characteristics
1	Klystron Signal Source	Sperry Gyroscope Co. Model 444
2	Klystron	Sperry Gyroscope Co. Type 2K43
3	Potentiometer Voltmeter	Bruno-New York Industries Model LVM-5
4	Frequency Meter	Narda Model 805 (500-1500 MHz)
5	Frequency Meter	Narda Model 806 (1500-2400 MHz)
6	Frequency Meter	Narda Model 802B (2350-10,500 MHz)
7	Frequency Meter	Hewlett-Packard Model X532A (8200-12,400 MHz)

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Item	Component	Type and Characteristics
8	Slotted line	Hewlett-Packard Model 850A (500-4000 MHz)
9	Slotted line and Probe Carriage	Hewlett-Packard Model 806B/809B (3000-12,000 MHz)
10	Standing wave indicator	Hewlett-Packard Model 415
11	RF Power meter	AN/USM-68 (Calibrated Standard)
12	Microwave Power Meter	Hewlett-Packard Model 430C
13	Thermistor Mount	Hewlett-Packard Model 477B
14	Variac 10A	General Radio Co.
15	AC Voltmeter	Weston Instruments Model 433
16	AC Vacuum Tube Volt Meter	Ballantine Laboratory Model 300
17	Polyrange meter	Sensitive Research Institute Corp. Model U
18	Decade Resistance Box	General Radio Co.
19	Attenuation Calibrator	Weinschel Engr Co. Model BA-5
20	RF Pulser	Weinschel Engr Co.
21	Audio Oscillator	Hewlett-Packard Co. Model 200 AB
22	Oscilloscope	Tektronix Inc. Model 531
23	Double Stub Tuner	Weinschel Engr Co. Model BS 109

4.5.12 Environmental test. The test set shall be subjected to the following environmental conditions in accordance with MIL-T-21200 with the following exceptions and additions. The over-all accuracy measurement listed in 4.5.8 shall be performed after each of the tests listed below. In addition, accuracy measurements are to be performed during the temperature and humidity tests. The accuracy of measurements performed during humidity and temperature testing shall not necessarily be within the specified limits (see 4.4); however, no degradation shall be caused in the performance (see 3.6) when subjected to the following conditions.

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4.5.12.1 Temperature test. The temperature test procedure shall be in accordance with MIL-STD-810 with the following exceptions:

a. Stabilize the test set at an ambient temperature of 25°C. Operate the test set for 20 minutes and measure performance. The performance shall not be below the minimum requirements specified herein.

b. With the test set still operating, lower the ambient temperature to minus 40°C and allow sufficient time for temperature stabilization. The performance shall not be below the minimum requirements specified herein and all controls and switches shall be readily manipulated.

c. With the test set still operating, slowly raise the ambient temperature to 52°C, and allow sufficient time for temperature stabilization; the performance shall not be below the minimum requirements specified herein.

4.5.12.2 Humidity test. The humidity test shall be as follows:

a. The test set shall be at a temperature of 55°C plus or minus 3°C and a relative humidity of not greater than 90 percent for a period of 24 hours.

b. Subject the test set to a minimum of 5 of the 48-hour cycles shown in MIL-STD-170.

4.5.12.3 Salt spray test. The contractor shall select, subject to the approval of the procuring activity, corrodible components and parts of the test set, and subject them to the salt spray test in accordance with MIL-STD-202, Method 101A. The duration of the test shall be 48 hours.

4.5.12.4 Watertightness test. The closed combination case shall show no evidence of water penetration when submerged in tap water, at room temperature to a minimum depth of 3 feet for a period of 4 hours.

4.5.13 Mechanical tests. The test set shall be subject to the following mechanical tests in accordance with MIL-T-4807:

a. Vibration: 1A

b. Shock: 2A

4.5.14 Reliability testing. The reliability of the equipment shall be tested as directed by the procuring activity in accordance with MIL-STD-785.

4.5.15 Interference test. The test set shall be tested for interference in accordance with MIL-STD-461.

4.6 Preproduction testing.

4.6.1 Preproduction test samples. The preproduction test samples shall consist of 2 models representative of the production equipment.

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4.6.2 Preproduction tests. The preproduction tests shall consist of all tests described under 4.5.

4.6.3 Preproduction test reports. When the contractor conducts the preproduction tests, he shall, after completion of the tests, prepare a production test report according to MIL-STD-831, and furnish three complete copies of the report to the procuring activity.

* 4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking for shipment and storage shall be examined and tested to determine compliance with the requirements of Section 5.

* 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order, the Test Set shall be preserved and packaged in accordance with MIL-P-116, Method IID. The combination case shall be used as the unit container and overboxed in a snug fitting container conforming to Grade W-5C, PPP-B-636. The desiccant, in quantities determined by Formula II of MIL-P-116, shall be secured inside the combination case in a manner that will not cause damage to the unit.

5.1.2 Level C. Test Sets shall be individually preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's standard practice, which may be used, provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Test Sets preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container shall be determined by the gross weight and size, and be limited to the special requirements of Table II of the specification. Waterproofing shall be in accordance with paragraph 30.4 of the appendix. Shipping containers conforming to PPP-B-601, overseas type, shall be used when the gross weight of the shipping container will exceed the weight limitations of PPP-B-636. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable.

5.2.2 Level B. Test Sets preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. The grade of each shipping container shall be determined by the gross weight and size and limited to the special requirements of Table I of the specification. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

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5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. Drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged and packed for immediate use or when drop tests of MIL-P-116 are applicable.

5.4 Marking. In addition to any other marking required by order or contract, the combination case, interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test set covered by this specification is intended for use as general purpose test equipment for measuring the rf power output of equipments operating within the frequency range of 500 MHz to 11,000 MHz, and with rf power outputs between 100 mw and 5 watts.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

c. Whether the testing will be conducted by the procuring activity or by the contractor (see 4.1).

d. Conditions under which reliability findings will be submitted for approval (see 3.3.2).

e. The quantity of preproduction samples to be tested (see 4.6.1).

f. Conditions under which Government-loaned property will be made available to the contractor (see 3.11).

* 6.3 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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